Dominic Rousseau (he/him)

github.com/itsdombo dominicrousseau.com contact@dominicrousseau.com (604) 618-3504

Skills

- **Programming Languages:** SystemVerilog, C/C++, Python, ARM Assembly, PLC Ladder Logic, Java, JavaScript
- **Software Tools:** Git, Docker, PlatformIO, Android Studio, Bash, TimescaleDB, Figma, Quartus
- **Libraries:** PyTorch, NumPy, Pandas, Matplotlib
- **Hardware:** Raspberry Pi, Arduino, ESP32, Embedded Systems Development, Soldering, Hardware Debugging

Education

University of British Columbia — *September 2022 - September 2028 (Planned)*
Bachelor of Applied Science, Computer Engineering

Work Experience

- **AI & Automation Programmer** - Greater Vancouver Food Truck Festivals — *May 2026 – Present*
  - Automated bookkeeper workflows using n8n, integrating Gmail, Google Sheets, Tally, and PostgreSQL on AWS Lightsail
  - Built an intake pipeline that parses booking emails, creates Tally forms via REST API, and connects vendors
  - Containerized the stack with Docker Compose and Caddy, using Cloudflare Tunnel for dev and Let's Encrypt for prod
- **Software Engineering Intern - Backend/APIs** - SteepDeep, Vancouver — *January 2026 - March 2026*
  - Designed a GraphQL-based API architecture to replace legacy REST endpoints using Firebase Data Connect
  - Built cloud-based backend services using Node.js and Google Cloud Platform for a SaaS application
  - Authored technical API documentation for third-party and frontend developer consumption in Github Projects

Engineering Design Team Experience

- **Captain** - UBC Biological Internet of Things (BIoT) — *May 2024 - April 2026*
  - Led 30 member team through full structural transition, ensuring continuous project development
  - Secured $5,500 in funding through successful grant applications for applied fermentation R&D project
- **Instrumentation Lead** - UBC Biological Internet of Things (BIoT) — *September 2022 - April 2024*
  - Introduced GitHub workflows
  - Built modular, low-cost instrumentation tools to monitor pH, temperature, and dissolved oxygen content
  - Networked ESP32s and Arduinos for real-time sensor data collection and display **Grafana** and **TimescaleDB**

Projects

- **Dynamic Autonomous Parking System (ROS2, PyTorch, YOLO, TensorRT, Jetson)** - Team Project
  - Built an autonomous parking system using ROS2, LiDAR, computer vision, and state-machine control
  - Trained a custom YOLO machine learning model on a self-labeled parking-sign dataset with transfer learning
  - Deployed real-time inference on NVIDIA Jetson Orin with TensorRT
- **Virtual Fridge Android App (Kotlin/Java) - Grocery Tracker** - Team Project
  - Deployed a full-stack Android application to AWS
  - Configured cloud infrastructure and environments to support scalable backend services using Docker
  - Integrated Google Gemini API for produce recognition
  - Established CI/CD pipelines on GitHub with Codacy, achieving a barcode retrieval in under 3 seconds
  - Utilized Jira for burndown charts and agile sprints
- **Cluck Guard** - Personal Project
  - Engineered a battery-powered door for a chicken coop, programmed in C++ and powered by a servo and photoresistor
  - Published comprehensive open-source build guide on GitHub, enabling replication and reuse by hobbyists and makers

Volunteer Experience

- **President** - Electrical and Computer Engineering Student Society — *May 2026 – Present*
- **Vice President, Student Life** - Electrical and Computer Engineering Student Society — *May 2025 – April 2026*
  - Organize orientations, game nights, and professional mixers for the department's undergraduate students
- **Server Infrastructure Manager** - Electrical and Computer Engineering Student Society — *November 2024 – Present*
  - Self-host and maintain a Minecraft server for 20 people, replacing paid third-party hosting
  - Automated boot, backup, and crash recovery processes using Bash scripting and systemd on Ubuntu
  - Manage network configuration including port forwarding, firewall rules, and DNS to ensure 24/7 secure remote access
- **Computer Electronics Repair Volunteer** - SPEC Repair Cafe — *February, 2023 - April 2024*
  - Fix and service electronic devices for members of the public